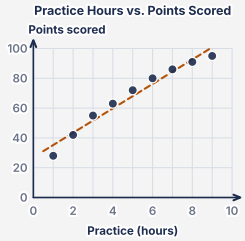
# Unit 7 Practice 1 — Data Analysis

Work through with your teacher · NC.8.SP.1–4

**ANSWER KEY**

**Describe correlation:** Up or down? Tight or spread out? Any outliers?

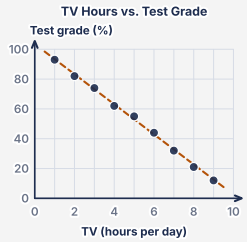
**Question 1** Score 3




Describe the correlation in this scatterplot.

- ● **Strong positive correlation**
- ○ Weak negative correlation
- ○ No correlation
- ○ Strong negative correlation

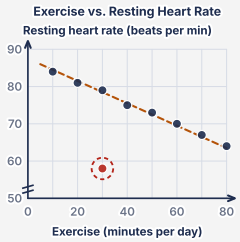
**Question 2** Score 3




What type of correlation does this scatterplot show?

- ○ Strong positive
- ○ Weak positive
- ● **Strong negative correlation**
- ○ No correlation

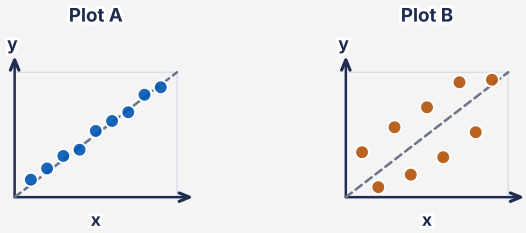
**Question 3** Score 4




The **red point** in the scatterplot is an outlier. What is the best description of its position?

- ○ It is the most accurate data point
- ● **Far from the trend — does not follow the overall pattern**
- ○ It is below the x-axis
- ○ It shows the average value

**Question 4** Score 4

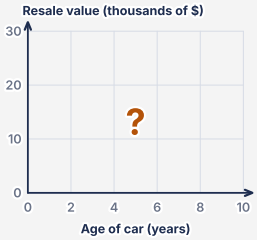



Both trend lines are the same. Look at how close the dots are to the line.

Which statement describes a *strong* correlation vs. a *weak* correlation?

- ○ Strong: the slope is steep · Weak: the slope is shallow
- ● **Strong: points cluster tightly around the trend line · Weak: points are spread far from the trend**
- ○ Strong: there are more data points · Weak: there are fewer points
- ○ Strong: the line goes up · Weak: the line goes down

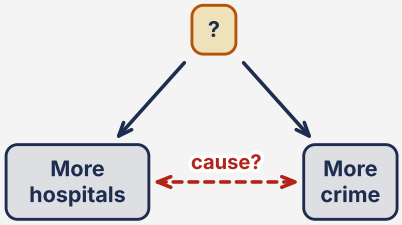
**Question 5** Score 4




A scatterplot for "age of car vs. resale value" would most likely show:

- ○ Strong positive correlation
- ● **Strong negative correlation**
- ○ No correlation
- ○ Perfect positive correlation

**Question 6** Score 5




Cities with more hospitals also have more crime. Could something else make both go up?

A student notices that cities with more hospitals also have higher crime rates. She concludes: "Hospitals cause crime." What is the flaw in her reasoning?

- ○ Her data must be wrong
- ● **Correlation does not imply causation — larger cities have both more hospitals AND more crime**
- ○ She is correct — the hospitals attract criminals
- ○ She needs a stronger scatterplot

**Question 7** Score 5



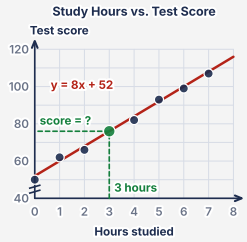

Lower reaction time = faster.

Two students both plot "hours of sleep vs. reaction time." Student A's dots fall in a tight line going down. Student B's dots show very little pattern. **Lower time = faster.** Which student's data best shows that more sleep makes reaction time faster?

- ○ Student B — the weak correlation is more convincing
- ● **Student A — strong negative correlation means more sleep → faster reaction (lower time)**
- ○ Student A — but only if the slope is positive
- ○ Student B — weak correlations are more reliable

**y = mx + b:** m = slope (rate of change per unit of x) · b = y-intercept (starting value when x = 0)

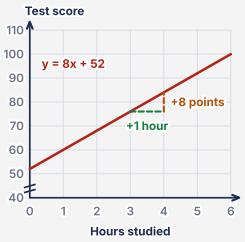
**Question 1** Score 3




Using y = 8x + 52: what score is predicted for a student who studies **3 hours**? (Find the green dot.)

- ○ 60 points
- ● **76 points**
- ○ 84 points
- ○ 52 points

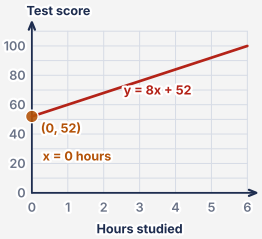
**Question 2** Score 3




In the context of *Study Hours vs. Test Score* (y = 8x + 52), what does the **slope of 8** mean?

- ○ A student starts with 8 points
- ● **For each additional hour of study, the predicted score increases by about 8 points**
- ○ The maximum possible score is 8
- ○ The score improves by 52 for each hour

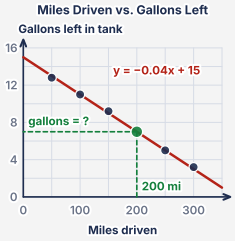
**Question 3** Score 4




In the same equation y = 8x + 52, what does the **y-intercept of 52** mean?

- ○ 52 is the highest possible score
- ● **A student who studies 0 hours is predicted to score about 52 points**
- ○ The student gains 52 points per hour
- ○ 52 hours of study is required to pass

**Question 4** Score 4




Using y = −0.04x + 15 (x = miles driven, y = gallons remaining): how many gallons remain after 200 miles? (Find the green dot.)

- ○ 11 gallons
- ● **7 gallons**
- ○ 15 gallons
- ○ 3 gallons

**Question 5** Score 5




Data was collected for x = 2 to x = 8. A student uses the equation to predict y at x = 25. Is x = 25 inside or outside the data? Is this prediction safe?

- ○ Inside the data — safe
- ● **Outside the data — less safe**
- ○ Inside the data — because x = 25 is a real number
- ○ Neither — predictions are always accurate

**Question 6** Score 5




The line of best fit is y = 5x + 10. Data was collected for x = 1 to x = 9. Which x-value is INSIDE the data, so it gives the safest prediction?

- ○ x = 15
- ● **x = 5 (inside the data range 1–9)**
- ○ x = 0
- ○ x = 20

**Joint frequency** = interior cells · **Marginal frequency** = row/column totals · **Relative frequency** = part ÷ a total. Use the grand total for everyone, or a row or column total for one group.

📋 Use this table for Questions 1–8 (Practice 1)

80 students were surveyed: Do you prefer **Morning** or **Evening** classes? Results split by class period:

| | Morning 🌅 | Evening 🌙 | Row Total |
|---|---|---|---|
| **Period 1–3** | **28** | 12 | **40** |
| **Period 4–6** | 18 | 22 | **40** |
| **Column Total** | **46** | **34** | **80** |

**Question 1** Score 3

| | Morning | Evening | Total |
|---|---|---|---|
| Period 1–3 | 28 | 12 | 40 |
| Period 4–6 | 18 | 22 | 40 |
| Total | 46 | 34 | 80 |

Find where the Period 1–3 row and the Morning column cross.

What is the **joint frequency** for Period 1–3 students who prefer Morning?

- ○ 40
- ● **28**
- ○ 46
- ○ 12

**Question 2** Score 3

| | Morning | Evening | Total |
|---|---|---|---|
| Period 1–3 | 28 | 12 | 40 |
| Period 4–6 | 18 | 22 | 40 |
| Total | 46 | 34 | 80 |

Marginal = the total at the edge of the Evening column.

What is the **marginal frequency** for all students who prefer Evening?

- ○ 22
- ● **34**
- ○ 12
- ○ 40

**Question 3** Score 4

| | Morning | Evening | Total |
|---|---|---|---|
| Period 1–3 | 28 | 12 | 40 |
| Period 4–6 | 18 | 22 | 40 |
| Total | 46 | 34 | 80 |

Period 1–3: $\frac{28}{40} = ?$

What percent of **Period 1–3** students prefer Morning? (Show calculation)

- ○ 35%
- ● **70%**
- ○ 46%
- ○ 57.5%

**Question 4** Score 4

| | Morning | Evening | Total |
|---|---|---|---|
| Period 1–3 | 28 | 12 | 40 |
| Period 4–6 | 18 | 22 | 40 |
| Total | 46 | 34 | 80 |

Period 4–6: $\frac{18}{40} = ?$

What percent of **Period 4–6** students prefer Morning?

- ○ 55%
- ○ 22.5%
- ● **45%**
- ○ 62.5%

**Question 5** Score 4

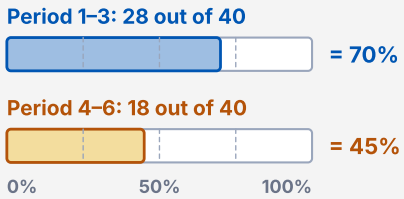
| | Morning | Evening | Total |
|---|---|---|---|
| Period 1–3 | 28 | 12 | 40 |
| Period 4–6 | 18 | 22 | 40 |
| Total | 46 | 34 | 80 |

Evening: $\frac{34}{80}$ = ?

What is the **relative frequency** of students who prefer Evening (out of 80 total)?

- ● **42.5%**
- ○ 27.5%
- ○ 55%
- ○ 22.5%

**Question 6** Score 5




Based on Questions 3 and 4 (70% vs 45%), is there an **association** between class period and morning preference?

- ○ No — both groups have the same preferences
- ● **Yes — Period 1-3 students strongly prefer Morning compared to Period 4-6**
- ○ No — the row totals are both 40
- ○ Yes — but only because the sample is small

**Question 7** Score 5

| | Morning | Evening | Total |
|---|---|---|---|
| Period 1–3 | 28 | 12 | 40 |
| Period 4–6 | 18 | 22 | 40 |
| Total | 46 | 34 | 80 |

The cell where Period 1–3 meets Evening, out of everyone (the grand total).

If a student is picked at random, what is the probability they are in Period 1-3 AND prefer Evening?

- ● **$^{12}/_{80}$ = 15%**
- ○ $^{34}/_{80}$ = 42.5%
- ○ $^{12}/_{40}$ = 30%
- ○ $^{22}/_{80}$ = 27.5%

## Study Tools

# Worked Examples

Click "Show Next Step" to reveal solutions one step at a time.

**Two-way table: 45 boys like pizza out of 120 total**

**Step 1: Joint relative freq**

```
45 ÷ 120 = 0.375 = 37.5%
```

| | Like pizza | Do not like | Total |
|---|---|---|---|
| Boys | 45 | 15 | 60 |
| Girls | | | 60 |
| Total | | | 120 |

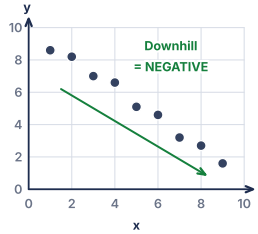
Joint: $\frac{45}{120} = 37.5\%$

Show Next Step →

## Common Mistakes

Can you spot what went wrong?

**Correlation: points go up-left to down-right**




✗ **Student:** Student says: positive correlation

✓ **Fix:** WRONG! Downhill = NEGATIVE correlation. Uphill = positive.

💡 *Think of it like a hill: going up = positive, going down = negative.*

## Key Tips for This Unit

✅ Positive: both increase. Negative: one up, other down. None: random.

✅ Line of best fit: approx equal points above and below.

✅ Outlier: far from the pattern. Can distort the trend line.

✅ Joint RF = cell÷grand total. Conditional RF = cell÷row or column total.